RUDRA PRATAP SINGH

AI/ML Engineer | LLMs, PyTorch Fine-Tuning, AI Agents and RAG

Bengaluru, Karnataka | rudra19867@gmail.com | linkedin.com/in/rudra-pratap-singh-5a0ab532a | github.com/rudraksh01-tech | Portfolio: rudraksh01-tech.github.io

SUMMARY

Computer Science student (B.E., CGPA 8.0) building Artificial Intelligence (AI), Machine Learning (ML) and Generative AI (GenAI) systems: PyTorch fine-tuning, LLM applications, AI agents and RAG, with measured results. Four live, tested projects; one cuts LLM calls by **57%** at near-equal accuracy. Seeking AI/ML Engineer and Machine Learning internships.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering (CSE) 2024 - 2028 (Expected) | 5th semester
Cambridge Institute of Technology, North Campus, Bengaluru | VTU Current CGPA: 8.0 / 10

Relevant coursework: Data Structures and Algorithms, OOP, Operating Systems, DBMS, Computer Networks

EXPERIENCE

Founding Software Engineer Intern, FERMOR | *Bengaluru, Karnataka (Hybrid)* Sep 2026 - Oct 2026

- Drove **about 30x** the usual daily traffic on launch day (a month's worth in one day) by redesigning key website pages end to end and optimizing them for Search Engine Optimization (SEO).
- Rebuilt the UI/UX with a cleaner frosted-glass design; tuned page structure, headings, metadata and content; used web analytics to measure the impact and plan next steps.

PROJECTS

Triage | LLM Cascade for Customer-Support Ticket Triage triage-rho-eight.vercel.app | Nov 2026
Python, PyTorch, Hugging Face, LoRA, scikit-learn, ONNX, FastAPI, Next.js, TypeScript, BullMQ | github.com/rudraksh01-tech/triage

- Cut LLM calls by **57%** at near-equal accuracy (**80.7%** vs **81.0%** LLM-only) by distilling 20,000 LLM labels into a fine-tuned PyTorch MiniLM classifier that labels a ticket in **8 ms** and escalates only low-confidence ones.
- Reduced calibration error from 0.108 to 0.066 (temperature scaling); benchmarked TF-IDF + logistic regression (63%) and LoRA (0.7% of parameters, same accuracy); quantized to ONNX int8, **4x faster**, after a parity check caught a silent export bug.
- Served **100 req/s** at 42 ms p95 with 0 errors via FastAPI behind a BullMQ/Redis queue, streaming to a Next.js/TypeScript inbox over WebSockets; a circuit breaker and graceful fallback kept it running when the LLM quota ran out.

Dwarpal | AI Security Gateway for LLM Agents (MCP) rudraksh01-tech.github.io/dwarpal | Nov 2026
Python, LangGraph, MCP, FastAPI, PostgreSQL, pgvector, Redis, Docker, Kubernetes, Helm | github.com/rudraksh01-tech/dwarpal

- Eliminated data leaks (**5 of 25** attack runs to **0 of 25**), with **0 false blocks** over 60 agent runs, via an open-source security gateway (guardrails) between LLM agents and MCP tool servers that blocks prompt injection, tool poisoning and rug pulls in English, Hindi, Hinglish and Kannada.
- Architected a LangGraph policy graph with human-in-the-loop approval: SHA-256 tool pinning, data-flow (taint) tracking, Llama Prompt Guard 2 and an LLM-as-a-judge with Redis-cached verdicts.
- Shipped Docker images and a Helm chart that CI/CD (GitHub Actions) deploys and tests on Kubernetes; PostgreSQL with pgvector for audit logs; **34** unit and integration tests.

AyurIP-Sahayak | Multilingual RAG Assistant ayur-ip-sahayak.vercel.app | Aug 2026 - Nov 2026
Python, FastAPI, React, Google Gemini API, NumPy, Docker, GitHub Actions | github.com/rudraksh01-tech/AyurIP-Sahayak

- Reached Hit@1 **0.95** and MRR **0.97** (vs 0.47 and 0.60 for BM25) with hybrid retrieval built from scratch (chunking, Gemini embeddings, BM25 + Reciprocal Rank Fusion) for a RAG assistant on Ayurveda and Indian IP law in 4 languages.
- Validated answers with an LLM-as-a-judge: **0.99** faithfulness (166 of 167 claims), 0.98 citation precision and **15 of 15** planted errors caught; **under 2 s** to first streamed words; 47 tests.

NestFinder | Agentic AI Assistant for Flat Search nestfinder-woad.vercel.app | Oct 2026 - Nov 2026
Node.js, Express.js, React, MongoDB, Groq API, Langfuse, OpenTelemetry | github.com/rudraksh01-tech/nestfinder

- Raised tool-choice accuracy from **18/20 to 20/20** with a 20-case agent evaluation of a tool-calling (function calling) AI agent built from scratch with streaming, structured JSON outputs and model fallback on rate limits.
- Instrumented LLM observability with OpenTelemetry and Langfuse (PII masked); added JWT authentication, rate limiting and MongoDB geospatial queries; **26** unit tests in CI; deployed on Vercel and Render.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL | **Frontend:** React (React.js), Next.js, Tailwind CSS, HTML, CSS

Machine Learning and Deep Learning: PyTorch, Hugging Face Transformers, Fine-Tuning, LoRA (PEFT), Knowledge Distillation, Text Classification, scikit-learn, Pandas, NumPy, Model Evaluation (Precision, Recall, F1), ONNX, Model Quantization, Model Serving

Generative AI (GenAI) and LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, LangGraph (LangChain), Model Context Protocol (MCP), Tool Calling, Prompt Engineering, Embeddings, Vector Databases, Semantic Search, Reranking, Natural Language Processing (NLP), LLM Evaluation, LLM-as-a-Judge, Guardrails

Backend and Data: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Pydantic, PostgreSQL, Redis, MongoDB, Message Queues (BullMQ), Rate Limiting, Caching

DevOps and Testing: Docker, Kubernetes, Helm, CI/CD (GitHub Actions), Git, Linux, Vercel, Render, Unit Testing (pytest), Playwright, Load Testing